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CHAPTER 1: LIMITS AND CONTINUITY

1.1 Definition of a Limit

Let f be a function defined on an open interval containing c (except possibly at c) and let L be a real number. The statement $\lim_{x \rightarrow c} f(x) = L$ means that for each $\epsilon > 0$ there exists a $\delta > 0$ such that for all x in the domain of f : if $0 < |x - c| < \delta$, then $|f(x) - L| < \epsilon$.

1.2 Limit Laws

Suppose $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$ exist.

1. Sum: $\lim[f(x) + g(x)] = L + M$
2. Product: $\lim[f(x) * g(x)] = L * M$
3. Quotient: $\lim[f(x) / g(x)] = L / M$, provided $M \neq 0$
4. Constant Multiple: $\lim[k * f(x)] = k * L$
5. Power: $\lim[f(x)]^n = L^n$

1.3 Squeeze Theorem

If $g(x) \leq f(x) \leq h(x)$ for all x in an open interval containing c (except possibly at c), and $\lim_{x \rightarrow c} g(x) = \lim_{x \rightarrow c} h(x) = L$, then $\lim_{x \rightarrow c} f(x) = L$.

Example: Find $\lim_{x \rightarrow 0} [\sin(x)/x]$

Since $-1 \leq \sin(x) \leq 1$ for all x , we can show that $\cos(x) \leq \sin(x)/x \leq 1$ for x near 0.

Since $\lim_{x \rightarrow 0} \cos(x) = 1$ and $\lim_{x \rightarrow 0} 1 = 1$, by the Squeeze Theorem, $\lim_{x \rightarrow 0} [\sin(x)/x] = 1$.

1.4 Continuity

A function f is continuous at $x = c$ if three conditions hold:

1. $f(c)$ is defined
2. $\lim_{x \rightarrow c} f(x)$ exists
3. $\lim_{x \rightarrow c} f(x) = f(c)$

Types of Discontinuity:

- Removable: limit exists but does not equal $f(c)$
- Jump: left and right limits exist but are different
- Infinite: one or both one-sided limits are infinite

CHAPTER 2: DERIVATIVES AND DIFFERENTIATION RULES

2.1 Definition of the Derivative

The derivative of f at $x = a$ is defined as:

$$f'(a) = \lim_{h \rightarrow 0} [f(a+h) - f(a)] / h$$

This represents the slope of the tangent line to the curve $y = f(x)$ at the point $(a, f(a))$.

2.2 Basic Differentiation Rules

Power Rule: $d/dx [x^n] = n * x^{(n-1)}$

Constant Rule: $d/dx [c] = 0$

Constant Multiple: $d/dx [cf(x)] = c * f'(x)$

Sum Rule: $d/dx [f(x) + g(x)] = f'(x) + g'(x)$

Product Rule: $d/dx [f(x)*g(x)] = f'(x)*g(x) + f(x)*g'(x)$

Quotient Rule: $d/dx [f/g] = [f'g - fg'] / g^2$

2.3 Chain Rule

If $y = f(u)$ and $u = g(x)$, then $dy/dx = f'(u) * g'(x)$.

Example: Find $d/dx [\sin(x^2)]$

Outer function: $\sin(u)$, derivative: $\cos(u)$

Inner function: $u = x^2$, derivative: $2x$

Result: $\cos(x^2) * 2x = 2x * \cos(x^2)$

2.4 Trigonometric Derivatives

$d/dx [\sin(x)] = \cos(x)$

$d/dx [\cos(x)] = -\sin(x)$

$d/dx [\tan(x)] = \sec^2(x)$

$d/dx [\sec(x)] = \sec(x)*\tan(x)$

$d/dx [\csc(x)] = -\csc(x)*\cot(x)$

$$d/dx [\cot(x)] = -\csc^2(x)$$

CHAPTER 3: APPLICATIONS OF DERIVATIVES

3.1 Related Rates

Steps for solving related rates problems:

1. Draw a diagram and identify given information
2. Define variables for all changing quantities
3. Write an equation relating the variables
4. Differentiate both sides with respect to time t
5. Substitute known values and solve

Example: A ladder 10 ft long rests against a vertical wall.

If the bottom slides away at 1 ft/s, how fast is the top sliding down when the bottom is 6 ft from the wall?

$$x^2 + y^2 = 100$$

$$2x(dx/dt) + 2y(dy/dt) = 0$$

$$\text{When } x = 6: y = 8, \text{ so } 12 + 16(dy/dt) = 0$$

$$dy/dt = -12/16 = -3/4 \text{ ft/s}$$

3.2 Optimization

Steps for optimization problems:

1. Identify the quantity to be optimized
2. Write the objective function
3. Express in terms of one variable using constraints
4. Find critical points: $f'(x) = 0$ or undefined
5. Test endpoints and critical points

Example: Find the dimensions of a rectangle with perimeter 40 ft that has maximum area.

$$\text{Let } x = \text{length, } y = \text{width. } 2x + 2y = 40, \text{ so } y = 20 - x$$

$$A(x) = x(20 - x) = 20x - x^2$$

$$A'(x) = 20 - 2x = 0 \rightarrow x = 10$$

$$A''(x) = -2 < 0, \text{ so } x = 10 \text{ is a maximum}$$

Dimensions: 10 ft by 10 ft (a square), Area = 100 sq ft

CHAPTER 4: INTEGRATION

4.1 Antiderivatives

An antiderivative of $f(x)$ is a function $F(x)$ such that $F'(x) = f(x)$.

The general antiderivative is $F(x) + C$ where C is a constant.

Basic Antiderivative Formulas:

$$\int x^n dx = x^{(n+1)}/(n+1) + C, n \neq -1$$

$$\int (1/x) dx = \ln|x| + C$$

$$\int e^x dx = e^x + C$$

$$\int \sin(x) dx = -\cos(x) + C$$

$$\int \cos(x) dx = \sin(x) + C$$

$$\int \sec^2(x) dx = \tan(x) + C$$

4.2 The Fundamental Theorem of Calculus

Part 1: If f is continuous on $[a,b]$ and $F(x) = \int(a \rightarrow x) f(t) dt$, then $F'(x) = f(x)$.

Part 2: If f is continuous on $[a,b]$ and F is any antiderivative of f , then $\int(a \rightarrow b) f(x) dx = F(b) - F(a)$.

Example: $\int(0 \rightarrow 1) (3x^2 + 2x) dx$

$$F(x) = x^3 + x^2$$

$$F(1) - F(0) = (1 + 1) - (0 + 0) = 2$$

CHAPTER 5: TECHNIQUES OF INTEGRATION

5.1 Substitution (u-substitution)

Let $u = g(x)$, then $du = g'(x) dx$.

$$\int f(g(x)) * g'(x) dx = \int f(u) du$$

Example: $\int 2x * \cos(x^2) dx$

$$\text{Let } u = x^2, du = 2x dx$$

$$\int \cos(u) du = \sin(u) + C = \sin(x^2) + C$$

5.2 Integration by Parts

$$\int u dv = uv - \int v du$$

Choose u using LIATE order:

Logarithmic, Inverse trig, Algebraic, Trig, Exponential

Example: $\int x * e^x dx$

$$u = x, dv = e^x dx$$

$$du = dx, v = e^x$$

$$= x * e^x - \int e^x dx = x * e^x - e^x + C$$

5.3 Partial Fractions

For proper rational functions $P(x)/Q(x)$ where $\deg(P) < \deg(Q)$:

Factor $Q(x)$ completely into linear and irreducible quadratic

Set up partial fraction decomposition

Solve for coefficients

Example: $\int (5x-4) / (x^2 - x - 2) dx$

$$x^2 - x - 2 = (x-2)(x+1)$$

$$(5x-4) / ((x-2)(x+1)) = A/(x-2) + B/(x+1)$$

$$5x - 4 = A(x+1) + B(x-2)$$

$$x = 2: 6 = 3A \rightarrow A = 2$$

$$x = -1: -9 = -3B \rightarrow B = 3$$

$$= \int [2/(x-2) + 3/(x+1)] dx = 2\ln|x-2| + 3\ln|x+1| + C$$